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পুর ও পরিবেশ কৌশল বিভাগ
শাহজালাল বিজ্ঞান ও প্রযুক্তি বিশ্ববিদ্যালয়
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SUB-SOIL INVESTIGATION REPORT

FOR PROPOSED ON 05 STORIED RESIDENTIAL BUILDING
AT PLOT # 129, WALI HOUSING ESTATE, BLOC-B, BOTESWAR, SYLHET
DAG # SA-1347, BS-4067, KHA # MUTATION-878, BS-924
J.L # SA-65, BS-57, MOUJA- BOTESWAR

CLIENT

MD. ABDUL WAHAB
S/O- LATE ABDUL HEKIM

Verification Code: rua939Zh
Verify report at: crtc-cee-sust.com

CONDUCTED BY

CENTER FOR RESEARCH TESTING & CONSULTANCY
DEPARTMENT OF CIVIL & ENVIRONMENTAL ENGINEERING
SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY, SYLHET

SAMPLE COLLECTED BY

THE UNIVERSAL SOIL & ENGINEERING
Moktosor Market (2nd Floor), Shop # 6, Bagbari Point, Sylhet.
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Introduction

A reasonably accurate conception about the sub-soil parameters of any project site is an essential prerequisite for proper planning and designing the foundation of the structure so that the same after its construction would remain safe and serviceable. A detailed study on the sub-surface elements in this regard would necessary to select proper type of foundation.

Objectives

The purpose of sub-soil investigation program is to obtain the following information

1. To classify the formation of soil layers
2. To estimate the safe bearing capacities of the foundation at different soil layers
3. To suggest the safe and appropriate type of foundations.

Scope of work

Centre for research, testing and consultancy (CRTC), Department of Civil and Environmental Engineering, Shahjalal University of Science & Technology (SUST), Sylhet; execute only the laboratory tests and other computational works with the supplied disturbed and/or undisturbed samples to determine the geotechnical properties of the soil.

Table – 1: Items of the supplied samples

No	Name	Bore hole no	Depth	Quantities
01	Disturbed sample	01	0-40 ft	08
02	Disturbed sample	02	0-40 ft	08
03	Disturbed sample	03	0-40 ft	08
04	Disturbed sample	04	0-40 ft	08
05	Disturbed sample	05	0-40 ft	08
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-

Laboratory Classifications

The disturbed soil samples, supplied by the client were thoroughly examined and classified roughly (color and type) and noted in accordance with bore hole (BH) and depth (Please see enclosed Tables).

Laboratory Works & Analysis

In addition to visual examination of the sample, Bearing Capacity of soil samples including skin friction & end bearing were calculated.

Bearing capacity for shallow foundation

The N value gives useful information with regards to consistency of cohesive soil and relative density of cohesion less soil. These relationships are given in the following Tables for cohesive and cohesion less soil, respectively.

Table 2: Relation of N-value with the Consistency and the Unconfined Compressive Strength of Cohesive Soils (after Bowles 1997)

N-values	Consistency	Unconfined Compressive Strength (q_u)	
		tsf	kPa
0-2	Very soft	0 to 0.25	0 to 25
3-5	Soft	0.25 to 0.50	25 to 50
6-9	Medium	0.50 to 1	50 to 100
10-16	Stiff	1 to 2	100 to 200
17-30	Very stiff	2 to 4	200 to 400
>30	Hard	>4	>400

Table 3: N and ϕ Relation to Relative Density for Fine Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
1 – 2	Very Loose	0 – 15	26 – 28	70 – 100
3 – 6	Loose	15 – 35	28 – 30	90 – 115
7 – 15	Medium	35 – 65	30 – 34	110 – 130
16 – 30	Dense	65 – 85	33 – 38	110 – 140
>30	Very Dense	>85	>50	130 – 150

Table 4: N and ϕ Relation to Relative Density for Medium Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
2 – 3	Very Loose	0 – 15	27 – 28	70 – 100
4 – 7	Loose	15 – 35	30 – 32	90 – 115
8 – 20	Medium	35 – 65	32 – 36	110 – 130
21 – 40	Dense	65 – 85	36 – 42	110 – 140

Table 5: N and ϕ Relation to Relative Density for Coarse Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
3 – 6	Very Loose	0 – 15	28 – 30	70 – 100
5 – 9	Loose	15 – 35	30 – 34	90 – 115
10 – 25	Medium	35 – 65	33 – 40	110 – 130
26 – 40	Dense	65 – 85	40 – 50	110 – 140
>40	Very Dense	>85	>50	130 – 150

<u>Bearing capacity for deep foundation</u>

Foundations, preferably piles are considered to transmit heavy load from superstructure to deep and hard strata. As the formation of soil strata consists of cohesive and non-cohesive type, the evaluation criteria for the determination of skin friction and the end bearing capacities of piles also vary. Details of the evaluation are provided in the following sub-headings.

Table 6: Formula Used to Determine Bearing Capacity of Soil

Shallow Foundation	
For Cohesive Soil	
$q_a = q_u + \gamma \frac{D}{3}$ $q_a = q_u$(Common Practice)	Where, q_a = Allowable bearing capacity, q_u = Unconfined Compressive Strength, γ = Unit Weight of soil, D = Depth of footing
For Cohesion less Soil	
$q_{ult} = 30N$; $q_a = q_{ult}/FS$	q_{ult} = Ultimate bearing capacity in kPa, N = SPT Number, FS = Factor of Safety
Source: Bowles, J. E., "Physical and Geotechnical Properties of Soils", 2nd Edition	
Deep Foundation	
For Cohesive Soil	
$f_{su} = \alpha C_u$; $q_{pu} = N_c C_u$	α = Adhesion factor, f_{su} = Ultimate skin friction, C_u = Undrained Shear Strength, q_{pu} = Ultimate end bearing of pile, N_c = Bearing capacity factor for deep foundation = 9
For Cohesion less Soil	
$f_{su} = N/50\ tsf$; $q_{pu} = 4N\ tsf$	f_{su} = Ultimate skin friction, q_{pu} = Ultimate end bearing of pile, N = SPT Number
Source: Rajapakse RA (2008), "Geotechnical Engineering Calculations and Rules of Thumb"	

Based on the above formula, the important parameters for foundation design have been determined and are given in **Tables 7-11**.

Table 7: The allowable bearing capacity for shallow foundation in Tsf (FS=2.5) **

Bore Hole No.	Depth	Design N	Clay						Sand
			Continuous			Square (Isolated)			
	(ft)	-	D/B=0.25	D/B=0.5	D/B=1	D/B=0.25	D/B=0.5	D/B=1	
BH01	2.5-5.0	5	0.603	0.634	0.692	0.712	0.770	0.837	0.650
	5.0-10	8	0.896	0.942	1.027	1.058	1.143	1.244	0.966
BH02	2.5-5.0	2	0.278	0.293	0.319	0.329	0.355	0.386	0.300
	5.0-10	8	0.908	0.955	1.041	1.072	1.158	1.260	0.978
BH03	2.5-5.0	5	0.580	0.610	0.665	0.685	0.740	0.805	0.625
	5.0-10	7	0.824	0.867	0.945	0.973	1.051	1.144	0.888
BH04	2.5-5.0	6	0.696	0.732	0.798	0.822	0.888	0.966	0.750
	5.0-10	13	1.460	1.536	1.674	1.724	1.863	2.026	1.573
BH05	2.5-5.0	5	0.534	0.561	0.612	0.630	0.681	0.741	0.575
	5.0-10	9	1.064	1.119	1.220	1.257	1.358	1.477	1.147
BH06	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH07	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH08	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH09	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH10	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-

**** Design engineer should consider the settlement issue along with the tabulated bearing capacity for designing the shallow foundation.**

Table 8: The allowable skin friction for Precast driven piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.049	0.049	0.118	0.138	0.118	-	-	-	-	-
10	0.138	0.074	0.096	0.118	0.096	-	-	-	-	-
15	0.138	0.049	0.138	0.156	0.118	-	-	-	-	-
20	0.239	0.144	0.239	0.200	0.144	-	-	-	-	-
25	0.218	0.074	0.118	0.174	0.118	-	-	-	-	-
30	0.136	0.200	0.214	0.218	0.261	-	-	-	-	-
35	0.224	0.224	0.224	0.208	0.224	-	-	-	-	-
40	0.216	0.216	0.216	0.216	0.216	-	-	-	-	-
45	-	-	-	-	-	-	-	-	-	-
50	-	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 9: The allowable end bearing for Precast driven piles in Tsf (FS = 2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.450	0.450	1.125	1.350	1.125	-	-	-	-	-
10	1.350	0.675	0.900	1.125	0.900	-	-	-	-	-
15	1.350	0.450	1.350	1.575	1.125	-	-	-	-	-
20	2.475	28.800	2.475	40.000	28.800	-	-	-	-	-
25	2.250	0.675	1.125	1.800	1.125	-	-	-	-	-
30	27.200	40.000	2.925	2.250	2.700	-	-	-	-	-
35	44.800	44.800	44.800	41.600	44.800	-	-	-	-	-
40	43.200	43.200	43.200	43.200	43.200	-	-	-	-	-
45	-	-	-	-	-	-	-	-	-	-
50	-	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 10: The allowable skin friction for Bored piles in Tsf (FS = 2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.032	0.032	0.077	0.090	0.077	-	-	-	-	-
10	0.090	0.048	0.062	0.077	0.062	-	-	-	-	-
15	0.090	0.032	0.090	0.101	0.077	-	-	-	-	-
20	0.156	0.094	0.156	0.130	0.094	-	-	-	-	-
25	0.141	0.048	0.077	0.113	0.077	-	-	-	-	-
30	0.088	0.130	0.139	0.141	0.170	-	-	-	-	-
35	0.146	0.146	0.146	0.135	0.146	-	-	-	-	-
40	0.140	0.140	0.140	0.140	0.140	-	-	-	-	-
45	-	-	-	-	-	-	-	-	-	-
50	-	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 11: The allowable end bearing for Bored piles in Tsf (FS = 2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.203	0.203	0.506	0.608	0.506	-	-	-	-	-
10	0.608	0.304	0.405	0.506	0.405	-	-	-	-	-
15	0.608	0.203	0.608	0.709	0.506	-	-	-	-	-
20	1.114	12.960	1.114	18.000	12.960	-	-	-	-	-
25	1.013	0.304	0.506	0.810	0.506	-	-	-	-	-
30	12.240	18.000	1.316	1.013	1.215	-	-	-	-	-
35	20.160	20.160	20.160	18.720	20.160	-	-	-	-	-
40	19.440	19.440	19.440	19.440	19.440	-	-	-	-	-
45	-	-	-	-	-	-	-	-	-	-
50	-	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

CONCLUSION AND RECOMMENDATION

On the basis of analysis and information provided by the owner following recommendations are suggested for the proposed five-storied residential building.

Soil profile scenario

Overall soil information of the investigated site is irregular in between the boreholes, and vertical inconsistency exists in the soil formation. In borehole 01, light brown soft clay with poor bearing capacity exist up to the top 25 ft. Then, grey fine sand with moderate to good bearing capacity are found. In borehole 02, light brown soft clay with poor bearing capacity is noticed up to the top 15 ft. After that grey fine sand, grey soft clay and grey fine sand with moderate to good bearing capacity are observed respectively. In borehole 03, light brown soft clay with poor bearing capacity available up to the top 25 ft. Then, grey stiff clay and grey fine sand with moderate to good bearing capacity are noticed in the further depth. In borehole 04, light brown soft clay with poor bearing capacity is observed up to the top 15 ft. After that grey fine sand, grey soft clay and grey fine sand with moderate to good bearing capacity are found respectively. Finally, in borehole 05, light brown soft clay with poor bearing capacity exist up to the top 15 ft. Then, grey fine sand, grey soft clay, grey stiff clay and grey fine sand with moderate to good bearing capacity are observed respectively.

Foundation options

In this situation, the foundation should be selected considering both the dimension and design load for the proposed building. Deep foundation (Bored/precast pile) may be selected to ensure the stability of the structure and to overcome the ground improvement complication. For designing any suitable deep foundation, the necessary data for skin friction and end bearing can be found from the attached table (8 to 11) in the report which is based on existing soil conditions.

Safety provisions

From the practical viewpoint, it has been found in many buildings that the structure failed due to soil settlement or judging the soil parameters incorrectly, providing no signs of structural damage. The earthquake vulnerability of the project area as well as the quality management of the construction work including workmanship should be taken into account for the safe design of the foundation.

Attachments

The layout of the boreholes of the investigated site and the bore-log profiles with SPT values is shown in the Appendix.

Special notes

These suggestions are absolutely tentative and the selection of type, depth, and dimensions of foundation would be decided by the Geotechnical Consultant or Design Engineer. It is also to be noted that the above recommendation is restricted only to the existing soil condition of 40 ft as the client provided.

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